Dimension-dependent minimax sensitivity for measuring quantum imaginarity

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We determine how much distinguishability a fixed quantum measurement can guarantee between an unknown state and its complex conjugate. The figure of merit is the worst-case ratio of classical total variation to trace-norm imaginarity. Optimizing over all POVMs gives an exact dimension-dependent constant, attained by a continuous covariant measurement on real orthonormal two-frames. The optimum scales as $d^{-1/2}$ and equals 3/4 for qutrits. On the same imaginary Hermitian sector, it exceeds the standard complex Haar measurement by the factor $d/(d-1)$. For finite measurements, we establish the exact outcome threshold for a positive guarantee and a dimension-explicit approximation bound. In qutrits we prove global optimality of the tetrahedral four-outcome POVM and give a five-outcome construction whose strict improvement is certified by outward interval arithmetic. Zonoid geometry supplies additional, nonmatching accuracy-dependent outcome bounds. Peak sensitivity and measurement-limited coherence conversion are treated as applications of the measurement-norm formulation.

I. INTRODUCTION

In a fixed reference basis, quantum imaginarity concerns the difference between a state and its entrywise complex conjugate. Resource-theoretic work has established quantification, state conversion, and distributed scenarios [1–4]. Here we ask a measurement-design question: if the input is unknown, what fraction of its conjugation distinguishability can one fixed POVM guarantee to retain in its classical outcomes? The reference basis and POVM are fixed before the input is chosen.

The appropriate mathematical language is that of measurement norms. Restricted-measurement distinguishability and its connection to quantum data hiding were developed by Matthews, Wehner, and Winter [13]. Aubrun and Lancien relate POVM distinguishability norms to zonoids and study sparsification of measurements [14]. We use that framework on the real vector space of purely imaginary Hermitian matrices. The two readout coefficients below are lower and upper domination constants on this subspace. We do not introduce a new general measurement norm. The optimization studied here ranges over all POVMs but only over conjugation-paired input directions; it is therefore different from restricting the measurement family for arbitrary state pairs.

Our principal result is the exact optimum of the lower domination constant in every finite dimension. The proof combines an upper bound from orthogonal conjugation of flat skew spectra with an attaining covariant POVM on real orthonormal two-frames. We also compute the standard complex Haar measurement on the same subspace. The relative improvement is exactly $d/(d-1)$: it is substantial in small dimension and tends to one as d increases. The dimension-dependent constant, its admissible input sector, and its attaining construction are the central objects of the paper.

For finite POVMs, we determine the exact minimum number of outcomes needed for a positive worst-case guarantee, and prove a randomized approximation bound of $O(d^2\epsilon^{-2}\log[2d/\delta])$ outcomes. In dimension three the optimum is 3/4; every finite-outcome POVM is strictly suboptimal. We prove that the tetrahedral measurement is globally optimal among four-outcome qutrit POVMs and give an explicit five-outcome improvement. Outward interval enclosures certify the latter comparison without relying on the displayed high-precision decimal alone. The zonoid accuracy bounds have different upper and lower exponents; the unrestricted five-outcome optimum also remains open.

Recent imaginarity studies address related but different tasks: coherence and discord conversion, measurement-assisted entanglement concentration, observable-based quantification, moment detection, and discrimination with imaginary auxiliary states under real measurements [5–12]. Measurement resource theories give the relevant postprocessing perspective [15]. Our coefficients describe the classical readout of a fixed POVM; they do not quantify all resources of an instrument or its implementation cost. A continuous POVM can be implemented by randomizing finite measurements [16], but a retained continuous random label is not a single finite outcome alphabet with the guarantee considered here.

Section II defines the subspace and readout constants. Section III proves the exact optimum and the Haar comparison. Section IV gives finite-outcome guarantees; Sec. V treats qutrit constructions. Peak sensitivity and coherence conversion are applications in Sec. VI. Section VII separates repeated-shot estimation from outcome complexity, and Sec. VIII summarizes the limitations.

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II. STATISTICAL SETTING AND MEASUREMENT NORMS

Fix a reference basis in dimension $d \geq 2$. Entrywise conjugation is denoted by ρ^*, and

$$\bar{\rho} = \frac{\rho + \rho^*}{2} \tag{1}$$

is a real density operator. We use the unnormalized trace norm and define

$$\mathcal{M}(\rho) = \min_{\sigma=\sigma^*} \|\rho - \sigma\|_1, \tag{2}$$

where the minimum is over real density operators.

Proposition 1. *For every state,*

$$\mathcal{M}(\rho) = \frac{1}{2}\|\rho - \rho^*\|_1. \tag{3}$$

Proof. For real σ, conjugation invariance and the triangle inequality give $\|\rho - \rho^*\|_1 \leq 2\|\rho - \sigma\|_1$. The choice $\sigma = \bar{\rho}$ attains half this distance.

For a finite POVM $\mathsf{E} = \{E_k\}$, let $p_k(\rho) = \mathrm{Tr}(E_k\rho)$ and $\mathrm{TV}[p,q] = \frac{1}{2}\sum_k |p_k - q_k|$. For measurable outcomes, total variation is $\sup_A |p(A) - q(A)|$. Define the guaranteed and peak coefficients, respectively, by

$$\underline{\mathfrak{I}_{\mathrm{meas}}}(\mathsf{E}) = \inf_{\mathcal{M}(\rho)>0} \frac{\mathrm{TV}[p(\rho), p(\rho^*)]}{\mathcal{M}(\rho)}, \tag{4}$$

$$\mathfrak{I}_{\mathrm{meas}}(\mathsf{E}) = \sup_{\mathcal{M}(\rho)>0} \frac{\mathrm{TV}[p(\rho), p(\rho^*)]}{\mathcal{M}(\rho)}. \tag{5}$$

Trace-distance contraction gives $0 \leq \underline{\mathfrak{I}_{\mathrm{meas}}} \leq \mathfrak{I}_{\mathrm{meas}} \leq 1$. Our main target is

$$\mathfrak{I}_d^{\mathrm{opt}} = \sup_{\mathsf{E}} \underline{\mathfrak{I}_{\mathrm{meas}}}(\mathsf{E}), \tag{6}$$

including continuous POVMs. This is a distinguishability coefficient, not a finite-copy minimax estimation risk. The definition does not assume a physical channel that conjugates an arbitrary unknown state.

Let $\mathcal{V}_d = \{X = X^\dagger : X^* = -X\}$ and, for finite outcomes, define

$$T_{\mathsf{E}}(X) = (\mathrm{Tr}\, E_1 X, \ldots, \mathrm{Tr}\, E_m X),$$
$$\|X\|_{\mathsf{E}} = \|T_{\mathsf{E}}(X)\|_{\ell_1}.$$

This is a seminorm, and it is a norm on $\mathcal{V}_d$ precisely when the readout map is injective there. If $X_\rho = \rho - \rho^*$, then

$$\frac{\mathrm{TV}[p(\rho), p(\rho^*)]}{\mathcal{M}(\rho)} = \frac{\|T_{\mathsf{E}}(X_\rho)\|_{\ell_1}}{\|X_\rho\|_1}. \tag{7}$$

Every nonzero $X \in \mathcal{V}_d$ is realizable up to scale by states $\mathbb{I}/d + tX/2$ for sufficiently small $t > 0$. Thus the two coefficients are exactly the lower and upper measurement-norm domination constants on $\mathcal{V}_d$ [13, 14]; in particular,

$$\mathfrak{I}_{\mathrm{meas}}(\mathsf{E}) = \|T_{\mathsf{E}}\|_{\mathcal{V}_d, \|\cdot\|_1 \to \ell_1}. \tag{8}$$

III. THE EXACT DIMENSION-DEPENDENT OPTIMUM

A. The optimal constant and an attaining POVM

Theorem 1 (Exact dimension-dependent minimax value)**.** *For $d \geq 2$, define*

$$\underline{\mathfrak{I}_{\mathrm{meas}}}(\mathsf{E}) = \inf_{\mathcal{M}(\rho)>0} \frac{\mathrm{TV}[p(\rho), p(\rho^*)]}{\mathcal{M}(\rho)},$$
$$\mathfrak{I}_d^{\mathrm{opt}} = \sup_{\mathsf{E}} \underline{\mathfrak{I}_{\mathrm{meas}}}(\mathsf{E}),$$

where the supremum is over all POVMs on a d-dimensional system, including continuous-outcome POVMs. Let

$$c_n = \frac{\Gamma(n/2)}{\sqrt{\pi}\,\Gamma((n+1)/2)} = \mathbb{E}_{\mathbf{u}\sim S^{n-1}}|u_1|.$$

Then

$$\mathfrak{I}_d^{\mathrm{opt}} = \begin{cases} c_{d-1}, & d \text{ even}, \\ \dfrac{d}{d-1} c_d, & d \text{ odd}. \end{cases} \tag{9}$$

The optimum is attained by the continuous POVM

$$\mathsf{E}(du\,dv) = d\,|\psi_{u,v}\rangle\langle\psi_{u,v}|\,d\nu(u,v), \quad |\psi_{u,v}\rangle = \frac{|u\rangle + i|v\rangle}{\sqrt{2}}, \tag{10}$$

where (u,v) ranges over ordered orthonormal pairs in $\mathbb{R}^d$ and ν is the invariant probability measure on this space.

Proof. Every imaginary Hermitian direction can, after a sufficiently small rescaling, be realized as $\rho - \rho^*$ for a density operator ρ near $\mathbb{I}/d$. Thus the minimax problem can be optimized over all nonzero matrices $X = X^\dagger$ with $X^* = -X$. Write a general POVM as $\mathsf{E}(d\omega) = F_\omega\, d\mu(\omega)$, where $F_\omega \geq 0$ and $\int \mathrm{Tr}\, F_\omega\, d\mu = d$. Its ratio on a direction X is

$$R_{\mathsf{E}}(X) = \frac{\int |\,\mathrm{Tr}(F_\omega X)|\, d\mu(\omega)}{\|X\|_1}.$$

We first prove the upper bound. For even d, choose A randomly from the invariant orbit of real orthogonal complex structures, so $A^T = -A$ and $A^T A = \mathbb{I}$. Then iA is Hermitian and $\|iA\|_1 = d$. For a normalized vector $|\psi\rangle = x + iy$, with $x, y \in \mathbb{R}^d$, one has

$$|\langle\psi|iA|\psi\rangle| = 2|x^T A y|.$$

For fixed y, the vector $Ay/\|y\|$ is uniformly distributed on the unit sphere in $y^\perp$. Consequently,

$$\mathbb{E}_A|\langle\psi|iA|\psi\rangle| \leq 2c_{d-1}\|x\|\,\|y\| \leq c_{d-1},$$

where the last inequality follows from $2\|x\|\|y\| \leq \|x\|^2 + \|y\|^2 = 1$. By spectral decomposition and the triangle inequality, every positive effect F obeys

$$\mathbb{E}_A|\,\mathrm{Tr}(FiA)| \leq c_{d-1}\,\mathrm{Tr}\, F.$$

Integrating over the POVM and using $\int \mathrm{Tr}\, F_\omega\, d\mu = d = \|iA\|_1$ gives $\mathbb{E}_A R_{\mathsf{E}}(iA) \le c_{d-1}$. Hence some input direction has ratio no larger than c_{d-1}.

For odd d, choose a random unit vector z, let A vanish on z, and let it act as an invariant orthogonal complex structure on $z^\perp$. Then $\|iA\|_1 = d-1$. For fixed nonzero y,

$$\|Ay\| = \|y\|\sqrt{1-(z\cdot\widehat{y})^2}.$$

The joint law is invariant under every orthogonal transformation fixing y. Therefore, conditional on its length, the direction of Ay is uniform in the unit sphere of $y^\perp$. Averaging this direction and then the length gives

$$\mathbb{E}_A|x^T Ay| = c_{d-1}a_d\|P_{y^\perp}x\|\,\|y\|.$$

The case $y=0$ contributes zero. For a normalized $\psi = x+iy$, this implies

$$\mathbb{E}_A|\langle\psi|iA|\psi\rangle| \le 2\|x\|\|y\|c_{d-1}a_d \le c_d,$$

where

$$a_d = \mathbb{E}_{\mathbf{z}\sim S^{d-1}}\sqrt{1-z_1^2} = \frac{\Gamma(d/2)^2}{\Gamma((d-1)/2)\Gamma((d+1)/2)}$$

and $c_{d-1}a_d = c_d$. Therefore $\mathbb{E}_A R_{\mathsf{E}}(iA) \le dc_d/(d-1)$, proving the odd-dimensional upper bound.

It remains to show attainability. The invariant average of the projectors in Eq. (10) is $\mathbb{I}/d$, so Eq. (10) defines a normalized POVM. Write $X = iA$, where the singular values of the real skew-symmetric matrix A, including zero pairs, are $s_1, s_1, \ldots, s_r, s_r$, with $r = \lfloor d/2\rfloor$ and one additional zero singular value when d is odd. Then $\|X\|_1/2 = \sum_j s_j$, and

$$R_{\mathsf{E}}(X) = \frac{d}{2\sum_j s_j}\,\mathbb{E}_{u,v}|u^T Av| = \frac{d\,c_{d-1}}{2\sum_j s_j}\,\mathbb{E}_u\|Au\|.$$

After putting A in its real canonical form, the last expectation is

$$F_d(\mathbf{s}) = \mathbb{E}_u\sqrt{\sum_{j=1}^{r} s_j^2(u_{2j-1}^2+u_{2j}^2)}.$$

For each fixed u, the integrand is a norm of $\mathbf{s}$; hence F_d is convex. The distribution of the paired coordinates is permutation-invariant, so F_d is symmetric and therefore Schur-convex. At fixed $\sum_j s_j$, it is minimized when all s_j are equal. For even d, this choice makes $\|Au\| = s$ for every unit u, giving $R_{\mathsf{E}}(X) = c_{d-1}$. For odd d, it gives $\|Au\| = s\sqrt{1-u_d^2}$ and $\mathbb{E}_u\|Au\| = sa_d$, so $R_{\mathsf{E}}(X) = dc_d/(d-1)$. Thus the constructed POVM has worst-case ratio equal to the upper bound, proving Eq. (9).

The first values are

d	2	3	4	5	6	7	8
$\mathfrak{I}_d^{\mathrm{opt}}$	1	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{15}{32}$	$\frac{3}{8}$	$\frac{35}{96}$	$\frac{5}{16}$

and Stirling's approximation gives $\mathfrak{I}_d^{\mathrm{opt}} \sim \sqrt{2/(\pi d)}$ as $d\to\infty$.

B. Comparison with the complex Haar measurement

Proposition 2 (A baseline on the same imaginary sector)**.** *Let $\mathsf{U}(d\psi) = d|\psi\rangle\langle\psi|\,d\mu(\psi)$, where μ is complex Haar probability measure on pure states. With $r = \lfloor d/2\rfloor$,*

$$\underline{\mathfrak{I}}_{\mathrm{meas}}(\mathsf{U}) = \frac{\Gamma(r+1/2)}{\sqrt{\pi}\,\Gamma(r+1)} = \frac{d-1}{d}\mathfrak{I}_d^{\mathrm{opt}}. \tag{11}$$

Proof. The Haar response on $X = iA$ is $d\mathbb{E}|\langle\psi|X|\psi\rangle|/\|X\|_1$. Its spectrum is $(s_1,-s_1,\ldots,s_r,-s_r)$, with one extra zero for odd d. At fixed $\sum_j s_j$, this expectation is a convex symmetric function of the nonnegative s_j. Averaging over permutations therefore shows that it is minimized by equal s_j.

In that case let p_+ and p_- be the Haar probabilities in the positive and negative r-dimensional eigenspaces. Conditional on $T = p_+ + p_-$, the fraction $U = p_+/T$ is Beta(r,r) and is independent of T; also $\mathbb{E}T = 2r/d$. Consequently the normalized response is

$$\frac{d}{2r}\mathbb{E}[T|2U-1|] = \mathbb{E}|2U-1|.$$

The beta integral gives

$$\mathbb{E}|2U-1| = \frac{2\,4^{-r}}{rB(r,r)} = \frac{\Gamma(r+1/2)}{\sqrt{\pi}\,\Gamma(r+1)}.$$

Comparison with Eq. (9) proves the final identity.

For qutrits the Haar value is $1/2$, while the optimum is $3/4$. More generally, the relative gain is $d/(d-1)$ and tends to one, so both measurements have the same leading $d^{-1/2}$ scaling. This baseline calculation concerns the same restricted input sector; it is not a comparison between different discrimination tasks.

C. The qutrit implementation and finite-outcome obstruction

The qutrit case of the general minimax theorem has a familiar spin-1 representation. Consider the Cartesian basis, where the three angular-momentum generators $\mathbf{J} = (J_x, J_y, J_z)$ are purely imaginary Hermitian matrices and satisfy $\mathrm{Tr}(J_iJ_j) = 2\delta_{ij}$. For a unit vector $\mathbf{n}\in S^2$, let $|\mathbf{n}\rangle$ be the spin-1 coherent state satisfying $(\mathbf{n}\cdot\mathbf{J})|\mathbf{n}\rangle = |\mathbf{n}\rangle$. The covariant POVM density

$$E(\mathbf{n}) = \frac{3}{4\pi}|\mathbf{n}\rangle\langle\mathbf{n}|, \qquad \int_{S^2} E(\mathbf{n})\,d\Omega = \mathbb{I}_3 \tag{12}$$

is normalized because $\int_{S^2}|\mathbf{n}\rangle\langle\mathbf{n}|\,d\Omega = (4\pi/3)\mathbb{I}_3$. For these continuous outcomes, $\mathrm{TV}[p,q] = \frac{1}{2}\int_{S^2}|p(\mathbf{n}) - q(\mathbf{n})|\,d\Omega$.

Theorem 2. *For every qutrit state ρ, the spin-1 covariant POVM in Eq. (12) satisfies*

$$\mathrm{TV}[p(\rho), p(\rho^*)] = \frac{3}{4}\mathcal{M}(\rho). \tag{13}$$

Because the ratio is constant for every nonzero imaginary direction in dimension three, both the peak and guaranteed sensitivities equal 3/4; the distinguishability bound is saturated for every input state.

Proof. In dimension three, every purely imaginary Hermitian matrix is a real linear combination of the spin-1 generators. Thus write $X = \rho - \rho^* = \mathbf{a}\cdot\mathbf{J}$. The eigenvalues of X are $0, \pm\|\mathbf{a}\|_2$, so $\mathcal{M}(\rho) = \|X\|_1/2 = \|\mathbf{a}\|_2$. The spin coherent state obeys $\langle\mathbf{n}|\mathbf{J}|\mathbf{n}\rangle = \mathbf{n}$, giving

$$p(\mathbf{n}|\rho) - p(\mathbf{n}|\rho^*) = \mathrm{Tr}[E(\mathbf{n})X] = \frac{3}{4\pi}\mathbf{a}\cdot\mathbf{n}.$$

Therefore,

$$\begin{aligned}\mathrm{TV}[p(\rho), p(\rho^*)] &= \frac{1}{2}\int_{S^2}\left|\frac{3}{4\pi}\mathbf{a}\cdot\mathbf{n}\right| d\Omega \\ &= \frac{3}{8\pi}\|\mathbf{a}\|_2\int_{S^2}|\cos\theta|\, d\Omega = \frac{3}{4}\|\mathbf{a}\|_2.\end{aligned}$$

This proves Eq. (13).

Theorem 3 (Finite-outcome obstruction to universal saturation)**.** *Let $d \geq 3$ and let E be a fixed finite-outcome POVM with $\mathfrak{I}_{\mathrm{meas}}(\mathsf{E}) > 0$. Then there is no equality*

$$\mathrm{TV}[p(\rho), p(\rho^*)] = \mathfrak{I}_{\mathrm{meas}}(\mathsf{E})\mathcal{M}(\rho)$$

for every input state ρ. Consequently, the coherence-conversion application in Sec. VI cannot be saturated for every input by one such finite-outcome protocol.

Proof. Restrict attention to a three-dimensional subspace and consider the imaginary Hermitian directions

$$X(\mathbf{a}) = i\begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0\end{pmatrix} \oplus 0_{d-3}, \qquad \mathbf{a}\in\mathbb{R}^3.$$

The eigenvalues of $X(\mathbf{a})$ are $0, \pm\|\mathbf{a}\|_2$, so $\|X(\mathbf{a})\|_1 = 2\|\mathbf{a}\|_2$. For sufficiently small $t > 0$, $\rho_t = \mathbb{I}/d + tX(\mathbf{a})/2$ is a state, $\rho_t - \rho_t^* = tX(\mathbf{a})$, and $\mathcal{M}(\rho_t) = t\|\mathbf{a}\|_2$. On the other hand,

$$\mathrm{TV}[p(\rho_t), p(\rho_t^*)] = \frac{t}{2}\sum_{k=1}^{m}|\,\mathrm{Tr}(E_k X(\mathbf{a}))|.$$

Each $\mathrm{Tr}(E_k X(\mathbf{a}))$ is a real linear function of $\mathbf{a}$. Thus the right-hand side is a finite sum of absolute values of linear functions, hence a polyhedral seminorm on $\mathbb{R}^3$. If universal saturation held with $\mathfrak{I}_{\mathrm{meas}}(\mathsf{E}) > 0$, this seminorm would equal $t\mathfrak{I}_{\mathrm{meas}}(\mathsf{E})\|\mathbf{a}\|_2$ in every direction. A nonzero Euclidean norm in three dimensions is not polyhedral, which is a contradiction. Finally, Eq. (31) is obtained through the chain $\Delta\mathcal{C} \leq \mathrm{TV} \leq \mathfrak{I}_{\mathrm{meas}}(\mathsf{E})\mathcal{M}$; equality at both endpoints for all states would force the forbidden universal equality in the middle.

The continuous qutrit measurement above has constant ratio, whereas a nonzero finite sum of absolute linear forms cannot equal that Euclidean norm. For qubits the imaginary sector is one-dimensional, and a fixed sharp Y measurement has ratio one for every non-real input.

Corollary 1 (Finite-outcome qutrit POVMs are strictly suboptimal)**.** *For every finite-outcome qutrit POVM, $\underline{\mathfrak{I}_{\mathrm{meas}}}(\mathsf{E}) < 3/4$.*

Proof. Theorem 1 gives $\underline{\mathfrak{I}_{\mathrm{meas}}}(\mathsf{E}) \leq 3/4$. If equality held, the ratio would be at least $3/4$ in every imaginary direction, while its spherical average is at most $3/4$; continuity would force the ratio to be constant. This contradicts the finite-outcome obstruction to universal saturation proved above.

IV. FINITE-OUTCOME GUARANTEES

Proposition 3 (Minimum number of outcomes for positive worst-case sensitivity)**.** *Let $\mathsf{E} = \{E_k\}_{k=1}^m$ be a finite-outcome POVM in dimension d, and let $p = d(d-1)/2$. If $m \leq p$, then*

$$\underline{\mathfrak{I}_{\mathrm{meas}}}(\mathsf{E}) = 0.$$

Conversely, there exists a POVM with exactly $p+1$ outcomes for which $\underline{\mathfrak{I}_{\mathrm{meas}}}(\mathsf{E}) > 0$. Thus the minimum number of outcomes required for strictly positive worst-case sensitivity is

$$m_{\min} = \frac{d(d-1)}{2} + 1.$$

Proof. The real vector space $\mathcal{V}_d = \{X = X^\dagger : X^* = -X\}$ has dimension p. Define the real-linear map

$$\begin{aligned}L_{\mathsf{E}} &: \mathcal{V}_d \longrightarrow \mathbb{R}^m, \\ L_{\mathsf{E}}(X) &= (\mathrm{Tr}(E_1 X), \ldots, \mathrm{Tr}(E_m X)).\end{aligned}$$

Because $\sum_k E_k = \mathbb{I}$ and every $X \in \mathcal{V}_d$ is traceless, the image of L_{E} lies in the $(m-1)$-dimensional hyperplane $\{y \in \mathbb{R}^m : \sum_k y_k = 0\}$. If $m \leq p$, rank-nullity gives a nonzero $X \in \ker L_{\mathsf{E}}$. Its normalized direction has zero measurement ratio, so $\underline{\mathfrak{I}_{\mathrm{meas}}}(\mathsf{E}) = 0$.

For attainability, choose a real basis $H_1, \ldots, H_p$ of $\mathcal{V}_d$, and set $H_{p+1} = -\sum_{j=1}^p H_j$. Let $m = p+1$ and $M = \max_{1\leq j\leq m}\|H_j\|_\infty$. Choose $0 < \eta < 1/(mM)$ and define

$$E_j = \frac{\mathbb{I}}{m} + \eta H_j, \qquad j = 1, \ldots, m.$$

Each H_j is traceless and Hermitian, $\sum_j H_j = 0$, and $\eta\|H_j\|_\infty < 1/m$, so every E_j is positive and $\sum_j E_j = \mathbb{I}$.

Hence these effects form a POVM. If $L_{\mathsf{E}}(X) = 0$, then $\mathrm{Tr}(H_j X) = 0$ for $j = 1, \ldots, p$, because $\mathrm{Tr}\, X = 0$. Since the H_j form a basis of $\mathcal{V}_d$ and the trace pairing is non-degenerate on this space, $X = 0$. Thus L_{E} is injective. By compactness of the trace-norm unit sphere in $\mathcal{V}_d$, the continuous function $\sum_j |\,\mathrm{Tr}(E_j X)|$ has a strictly positive minimum on that sphere. Therefore $\underline{\mathfrak{I}}_{\mathrm{meas}}(\mathsf{E}) > 0$, proving the converse.

Theorem 4 (Improved finite-outcome approximation bound)**.** *Let* $\mathfrak{I}_d^{\mathrm{opt}}$ *be the optimum in Theorem 1, and choose* $0 < \epsilon, \delta < 1$*. Set*

$$t = \frac{\epsilon \mathfrak{I}_d^{\mathrm{opt}}}{4d}, \qquad \kappa = \frac{\epsilon \mathfrak{I}_d^{\mathrm{opt}}}{64}.$$

There is a universal constant $C > 0$ *such that the following conditions are sufficient:*

$$N \geq C \left[\frac{\log(2d)}{dt^2} + \frac{\log(2d)}{t} \right], \tag{14}$$

$$N \geq \frac{32 \log(2/\delta)}{d^2 t^2}, \tag{15}$$

$$N \geq \frac{8 \log(2/\delta)}{t}, \tag{16}$$

$$N \geq \frac{2(d-1) + \frac{2}{3} d\kappa}{\kappa^2} \log \frac{4d}{\delta}. \tag{17}$$

Then, with probability at least $1-\delta$*, there is a normalized* N*-outcome POVM* F_N *such that*

$$\underline{\mathfrak{I}}_{\mathrm{meas}}(\mathsf{F}_N) \geq (1-\epsilon)\mathfrak{I}_d^{\mathrm{opt}}. \tag{18}$$

Since $\mathfrak{I}_d^{\mathrm{opt}} = \Theta(d^{-1/2})$*, these conditions imply the asymptotic upper bound*

$$N = O\left(d^2 \epsilon^{-2} \log \frac{2d}{\delta} \right).$$

No matching lower bound is claimed.

Proof. Draw N independent invariant orthonormal frames (u_j, v_j) and set $|\psi_j\rangle = (|u_j\rangle + i|v_j\rangle)/\sqrt{2}$, $P_j = |\psi_j\rangle\langle\psi_j|$, and $E_j^{(0)} = (d/N)P_j$. The effects $E_j^{(0)}$ are positive but need not sum exactly to the identity. Let $S = \sum_j E_j^{(0)}$ and, whenever S is invertible, define

$$F_j = S^{-1/2} E_j^{(0)} S^{-1/2}.$$

Then $\sum_j F_j = \mathbb{I}$, so $\mathsf{F}_N = \{F_j\}_{j=1}^N$ is a genuine POVM.

We first control the sampling error uniformly over all imaginary directions. Write $X = iA$, where A is real skew-symmetric and $\|A\|_1 = \|X\|_1 = 1$. The continuous frame POVM has ratio $d\,\mathbb{E}|u^T A v|$, whereas the unnormalized sampled effects have ratio $dN^{-1}\sum_{j=1}^N |u_j^T A v_j|$. Define

$$D_N = \sup_{\substack{A^T = -A \\ \|A\|_1 = 1}} \left| \frac{1}{N} \sum_{j=1}^N |u_j^T A v_j| - \mathbb{E}|u^T A v| \right|.$$

The random variables in this empirical process lie in $[0, 1/2]$, since the nonzero singular values of a real skew-symmetric matrix occur in equal pairs and hence $\|A\|_\infty \leq \|A\|_1/2 = 1/2$. By homogeneity, the supremum defining D_N is unchanged if the constraint $\|A\|_1 = 1$ is replaced by $\|A\|_1 \leq 1$, so the index set can be taken to contain the zero matrix.

For each frame define the real skew-symmetric matrix $W_j = (u_j v_j^T - v_j u_j^T)/2$, so that $u_j^T A v_j = \mathrm{Tr}(A^T W_j)$. Symmetrization and the contraction inequality for Rademacher averages [17] give

$$\mathbb{E} D_N \leq \frac{C_1}{N}\, \mathbb{E} \left\| \sum_{j=1}^N \varepsilon_j W_j \right\|_\infty,$$

where ε_j are independent random signs and C_1 is a universal constant. The matrices $i\varepsilon_j W_j$ are centered independent Hermitian matrices, satisfy $\|iW_j\|_\infty = 1/2$, and obey

$$\mathbb{E}[(iW_j)^2] = \frac{\mathbb{I}}{2d}.$$

The expected-norm form of matrix Bernstein concentration [18] therefore yields

$$\mathbb{E} \left\| \sum_{j=1}^N \varepsilon_j W_j \right\|_\infty \leq C_2 \left[\sqrt{\frac{N \log(2d)}{d}} + \log(2d) \right],$$

with a universal constant C_2. Consequently,

$$\mathbb{E} D_N \leq C_3 \left[\sqrt{\frac{\log(2d)}{Nd}} + \frac{\log(2d)}{N} \right].$$

Choosing the universal constant in Eq. (14) sufficiently large makes $\mathbb{E} D_N \leq t/4$.

For a uniform orthonormal frame, conditioning on u and averaging over $v \perp u$ gives

$$\mathbb{E}|u^T A v|^2 = \frac{\|A\|_F^2}{d(d-1)} \leq \frac{1}{2d(d-1)} \leq \frac{1}{d^2}.$$

Here $\|A\|_F^2 \leq \|A\|_1^2/2 = 1/2$ because the singular values of a real skew-symmetric matrix occur in equal pairs. Apply Bousquet's variance-sensitive inequality for suprema of empirical processes [19] to the class consisting of $f_A(u, v) = |u^T A v|$ and $1/2 - f_A(u, v)$. Both functions take values in $[0, 1/2]$, and their empirical deviations cover both signs in D_N. With $x = \log(2/\delta)$ and $\sigma^2 \leq d^{-2}$, the bound reads

$$\Pr\left[D_N \geq \mathbb{E} D_N + \sqrt{2\left(\frac{\sigma^2 x}{N} + \frac{x\, \mathbb{E} D_N}{N} \right)} + \frac{x}{6N} \right] \leq e^{-x}.$$

Conditions (15) and (16), together with $\mathbb{E} D_N \leq t/4$, make the square-root term at most $\sqrt{2}t/4$ and the final term at most $t/48$. Hence $D_N \leq t$ except on an event

of probability at most $\delta/2$. It follows that the unnormalized empirical ratio differs from the continuous ratio by at most $dt = \epsilon\mathfrak{I}_d^{\mathrm{opt}}/4$, uniformly over all imaginary directions.

It remains to normalize the sampled effects. Since $\mathbb{E}P_j = \mathbb{I}/d$,

$$S - \mathbb{I} = \frac{1}{N}\sum_{j=1}^{N}(dP_j - \mathbb{I}).$$

The summands are independent, centered Hermitian matrices with operator norm at most d and variance parameter $d-1$. Matrix Bernstein concentration [18] and Eq. (17) imply $\|S-\mathbb{I}\|_\infty \le \kappa$ except with probability at most $\delta/2$. In particular, S is invertible. Put $B = S^{-1/2}$. For $\kappa \le 1/2$, one has $\|B\|_\infty \le \sqrt{2}$ and $\|B-\mathbb{I}\|_\infty \le 2\kappa$. For every Hermitian X with $\|X\|_1 = 1$,

$$\begin{aligned}
&\left|\sum_j |\operatorname{Tr}(F_jX)| - \sum_j |\operatorname{Tr}(E_j^{(0)}X)|\right| \\
&\quad\le \sum_j |\operatorname{Tr}[E_j^{(0)}(BXB - X)]| \\
&\quad\le \|S\|_\infty \|BXB - X\|_1 \\
&\quad\le 8\kappa = \frac{\epsilon\mathfrak{I}_d^{\mathrm{opt}}}{8}.
\end{aligned}$$

Combining the two events, the total deviation from the continuous optimum is at most $3\epsilon\mathfrak{I}_d^{\mathrm{opt}}/8$, which is smaller than $\epsilon\mathfrak{I}_d^{\mathrm{opt}}$. Therefore Eq. (18) holds with probability at least $1-\delta$.

Theorem 5 (Qutrit outcome complexity near the minimax optimum)**.** *Let $m_3(\epsilon)$ be the smallest number of outcomes of a qutrit POVM E satisfying*

$$\underline{\mathfrak{I}}_{\mathrm{meas}}(\mathsf{E}) \ge (1-\epsilon)\frac{3}{4}.$$

There are constants $c, C > 0$, independent of $0 < \epsilon \le 1/2$, such that

$$c\,\epsilon^{-2/5} \le m_3(\epsilon) \le C\,\epsilon^{-4/5}. \tag{19}$$

The upper bound is an existence result obtained by discretizing the spin-1 covariant POVM and then correcting its normalization. The lower bound applies to every finite-outcome qutrit POVM. The exponents do not match, so Eq. (19) does not determine the exact accuracy dependence.

Proof. In the spin-1 representation, write $X = \mathbf{a}\cdot\mathbf{J}$ with $\|\mathbf{a}\|_2 = 1$, so $\|X\|_1 = 2$. For a finite POVM define

$$\mathbf{b}_k = \big(\operatorname{Tr}(E_kJ_x), \operatorname{Tr}(E_kJ_y), \operatorname{Tr}(E_kJ_z)\big).$$

Its directional response is

$$h(\mathbf{a}) = \frac{1}{2}\sum_{k=1}^{m}|\mathbf{a}\cdot\mathbf{b}_k|.$$

This is the support function of the zonotope

$$Z = \sum_{k=1}^{m}[-\mathbf{b}_k/2, \mathbf{b}_k/2] \subset \mathbb{R}^3.$$

Positivity of each effect and $\|\mathbf{n}\cdot\mathbf{J}\|_\infty = 1$ imply $\|\mathbf{b}_k\|_2 \le \operatorname{Tr}E_k$, so $\sum_k \|\mathbf{b}_k\|_2 \le 3$. Averaging over the unit sphere gives

$$\int_{S^2} h(\mathbf{a})\,d\sigma(\mathbf{a}) = \frac{1}{4}\sum_k \|\mathbf{b}_k\|_2 \le \frac{3}{4},$$

where σ is normalized spherical measure.

For the upper bound, let σ itself be the uniform probability measure on S^2. A theorem on zonoid approximation states that for every integer n there is a probability measure μ_n supported on at most n points such that

$$\sup_{\mathbf{a}\in S^2}\left|\int|\mathbf{a}\cdot\mathbf{n}|\,d\mu_n(\mathbf{n}) - \frac{1}{2}\right| \le C_0 n^{-5/4}$$

for a constant C_0 depending only on the fixed sphere dimension [20].

The probability normalization does not change this rate. If the approximation theorem is stated for a finite positive generating measure of mass M, integrating its cosine transform over the sphere gives $|M-1| \le 2\delta$ when the uniform error is δ. Dividing the measure by M changes the error to at most $2\delta/(1-2\delta) \le 4\delta$ for $\delta \le 1/4$. This factor is absorbed into C_0. The quoted approximation input is for the ambient three-dimensional ball, with directions on S^2.

Replace μ_n by its antipodally symmetrized measure; this at most doubles the support and leaves the displayed cosine transform unchanged. Write the resulting measure as $\mu = \sum_{j=1}^{M} w_j\delta_{\mathbf{n}_j}$, where $M \le 2n$, and set

$$E_j^{(0)} = 3w_jP_{\mathbf{n}_j}, \qquad P_{\mathbf{n}} = |\mathbf{n}\rangle\langle\mathbf{n}|.$$

The effects are positive but need not sum exactly to $\mathbb{I}_3$. In the Cartesian spin-1 representation, $P_{\mathbf{n}}+P_{-\mathbf{n}} = \mathbb{I}_3 - \mathbf{n}\mathbf{n}^T$. Let $Q = \int \mathbf{n}\mathbf{n}^T\,d\mu(\mathbf{n})$. Antipodal symmetry then gives

$$S := \sum_j E_j^{(0)} = \frac{3}{2}(\mathbb{I}_3 - Q).$$

The cosine-transform error controls the second moment. Indeed, rotational symmetry gives

$$\int_{S^2}|\mathbf{a}\cdot\mathbf{n}|\left(\mathbf{a}\mathbf{a}^T - \frac{\mathbb{I}_3}{3}\right)d\sigma(\mathbf{a}) = \frac{1}{8}\left(\mathbf{n}\mathbf{n}^T - \frac{\mathbb{I}_3}{3}\right).$$

Consequently, if the uniform cosine-transform error is at most δ, then

$$\left\|Q - \frac{\mathbb{I}_3}{3}\right\|_\infty \le \frac{16}{3}\delta, \qquad \|S - \mathbb{I}_3\|_\infty \le 8\delta.$$

Choose n large enough that $C_0 n^{-5/4} \le \delta$, and take $\delta = \epsilon/100$. Then S is invertible. Define

$$F_j = S^{-1/2} E_j^{(0)} S^{-1/2}.$$

These effects form an exact POVM. Before normalization, its response differs from $3/4$ by at most $3\delta/2$ uniformly in $\mathbf{a}$. If $\kappa = \|S - \mathbb{I}_3\|_\infty \le 8\delta \le 1/2$, the whitening correction changes the response by at most $8\kappa \le 64\delta$, using trace/operator-norm duality. Thus

$$\underline{\mathfrak{I}}_{\text{meas}}(\mathsf{F}) \ge \frac{3}{4} - \frac{131}{2}\delta \ge \frac{3}{4}(1-\epsilon).$$

Since $n = O(\delta^{-4/5}) = O(\epsilon^{-4/5})$ and $M \le 2n$, this proves the upper bound.

For the lower bound, suppose $\underline{\mathfrak{I}}_{\text{meas}}(\mathsf{E}) \ge (1-\epsilon)3/4$. Let

$$K = \frac{Z}{(1-\epsilon)3/4}$$

and let H be its support function. Then $H(\mathbf{a}) \ge 1$ for all $\mathbf{a} \in S^2$, while

$$\int_{S^2} H(\mathbf{a})\, d\sigma(\mathbf{a}) \le \frac{1}{1-\epsilon}.$$

Writing $K = \sum_k [-\mathbf{g}_k/2, \mathbf{g}_k/2]$ with $\mathbf{g}_k = \mathbf{b}_k/[(1-\epsilon)(3/4)]$, the sum of the half-lengths is bounded by

$$\frac{1}{2}\sum_k \|\mathbf{g}_k\|_2 = \frac{\sum_k \|\mathbf{b}_k\|_2}{2(1-\epsilon)(3/4)} \le \frac{2}{1-\epsilon}.$$

Consequently $H(\mathbf{a}) \le 2/(1-\epsilon) \le 4$. Put $\Delta = \max_{\mathbf{a}\in S^2} H(\mathbf{a}) - 1$. For $\epsilon \le 1/2$, this gives $\Delta \le 3$.

Let $\mathbf{a}_0$ maximize H, and choose a support point $v_0 \in K$ with $\mathbf{a}_0 \cdot v_0 = H(\mathbf{a}_0) = 1 + \Delta$. Since $\max_{v\in K} \|v\| = \max_{\mathbf{a}} H(\mathbf{a}) = 1 + \Delta$, equality in Cauchy–Schwarz forces $v_0 = (1+\Delta)\mathbf{a}_0$. Hence

$$H(\mathbf{a}) \ge \mathbf{a} \cdot v_0 = (1+\Delta)\mathbf{a}_0 \cdot \mathbf{a}.$$

This support-point argument does not require differentiability of the zonotope support function.

On the spherical cap defined by $\mathbf{a}_0 \cdot \mathbf{a} \ge (1+\Delta/2)/(1+\Delta)$, one therefore has $H(\mathbf{a}) \ge 1 + \Delta/2$. The normalized area of this cap is $\Delta/[4(1+\Delta)]$, so

$$\frac{\Delta^2}{8(1+\Delta)} \le \int_{S^2} (H-1)\, d\sigma \le \frac{\epsilon}{1-\epsilon}.$$

Using $\Delta \le 3$ and $\epsilon \le 1/2$ yields $\Delta \le 8\sqrt{\epsilon}$.

The zonotope approximation lower bound for the three-dimensional Euclidean ball states that any zonotope with at most m generating segments satisfying $B_2^3 \subseteq K \subseteq (1+\Delta)B_2^3$ must have $\Delta \ge c_0 m^{-5/4}$ for a dimension-dependent constant $c_0 > 0$ [21]. Combining this with $\Delta \le 8\sqrt{\epsilon}$ gives $m \ge (c_0/8)^{4/5}\epsilon^{-2/5}$, proving the lower bound.

V. QUTRIT CONSTRUCTIONS AND EXACT BENCHMARKS

Theorem 6 (Global optimality of the tetrahedral four-outcome qutrit POVM)**.** *Among all qutrit POVMs with exactly four outcomes,*

$$\sup_{\mathsf{E}:\,|\mathsf{E}|=4} \underline{\mathfrak{I}}_{\text{meas}}(\mathsf{E}) = \sqrt{\frac{3}{8}}. \tag{20}$$

The supremum is attained by the tetrahedral POVM $E_i = \frac{3}{4}P_{\mathbf{n}_i}$, where the $\mathbf{n}_i$ are the four unit vertices of a regular tetrahedron and $P_{\mathbf{n}_i}$ is the spin-1 coherent projector.

Proof. In the Cartesian spin-1 representation, let $\mathbf{J} = (J_x, J_y, J_z)$ be the three imaginary Hermitian generators, normalized so that $\|\mathbf{a}\cdot\mathbf{J}\|_\infty = \|\mathbf{a}\|_2$ and $\|\mathbf{a}\cdot\mathbf{J}\|_1 = 2\|\mathbf{a}\|_2$. For a four-outcome POVM $\mathsf{E} = \{E_i\}_{i=1}^4$, define vectors $\mathbf{b}_i \in \mathbb{R}^3$ by

$$\mathbf{b}_i \cdot \mathbf{a} = \operatorname{Tr}[E_i(\mathbf{a}\cdot\mathbf{J})].$$

Since $\sum_i E_i = \mathbb{I}_3$ and $\operatorname{Tr} J_\alpha = 0$, these vectors satisfy $\sum_i \mathbf{b}_i = 0$. Positivity gives $\|\mathbf{b}_i\|_2 \le \operatorname{Tr} E_i$, and hence $L := \sum_i \|\mathbf{b}_i\|_2 \le 3$. Writing B for the 4×3 matrix whose rows are $\mathbf{b}_i^T$, the worst-case sensitivity is

$$\underline{\mathfrak{I}}_{\text{meas}}(\mathsf{E}) = \frac{1}{2}\min_{\|\mathbf{a}\|_2=1} \|B\mathbf{a}\|_1.$$

If $\operatorname{rank} B < 3$, this quantity is zero. Suppose therefore that $\operatorname{rank} B = 3$, and put $G = B^T B$. The column space of B is the three-dimensional subspace $H = \{\mathbf{y} \in \mathbb{R}^4 : \sum_i y_i = 0\}$, so $BB^+ = P_H$, where $B^+ = G^{-1}B^T$ and $P_H = \mathbb{I}_4 - \frac{1}{4}\mathbf{1}\mathbf{1}^T$.

For each $i < j$, set $\mathbf{y}_{ij} = (\mathbf{e}_i - \mathbf{e}_j)/\sqrt{2}$. These six unit vectors lie in H, satisfy $\|\mathbf{y}_{ij}\|_1 = \sqrt{2}$, and obey

$$\sum_{i<j} \mathbf{y}_{ij}\mathbf{y}_{ij}^T = 2P_H.$$

Choosing $\mathbf{a} = B^+\mathbf{y}_{ij}/\|B^+\mathbf{y}_{ij}\|_2$ gives $B\mathbf{a} = \mathbf{y}_{ij}/\|B^+\mathbf{y}_{ij}\|_2$. Consequently,

$$\min_{\|\mathbf{a}\|_2=1} \|B\mathbf{a}\|_1 \le \frac{\sqrt{2}}{\max_{i<j}\|B^+\mathbf{y}_{ij}\|_2} \le \sqrt{\frac{6}{\operatorname{Tr} G^{-1}}}.$$

To bound the last expression, let $\ell_i = \|\mathbf{b}_i\|_2$ and $\mathbf{u}_i = G^{-1}\mathbf{b}_i$. The diagonal entries of $BB^+ = P_H$ imply $\mathbf{b}_i^T G^{-1}\mathbf{b}_i = 3/4$, so $\ell_i\|\mathbf{u}_i\|_2 \ge 3/4$. It follows that

$$\operatorname{Tr} G^{-1} = \sum_i \|\mathbf{u}_i\|_2^2 \ge \frac{9}{16}\sum_i \frac{1}{\ell_i^2}.$$

By Cauchy–Schwarz, $(\sum_i \ell_i)(\sum_i \ell_i^{-1}) \ge 16$ and $\sum_i \ell_i^{-2} \ge \frac{1}{4}(\sum_i \ell_i^{-1})^2$. Therefore

$$L^2 \operatorname{Tr} G^{-1} \ge \frac{9}{16}L^2 \sum_i \ell_i^{-2} \ge 36.$$

Combining the bounds yields

$$\underline{\mathfrak{I}}_{\mathrm{meas}}(\mathsf{E}) \le \frac{L}{2\sqrt{6}} \le \frac{3}{2\sqrt{6}} = \sqrt{\frac{3}{8}}.$$

For the tetrahedral POVM, the effects sum to $\mathbb{I}_3$ and $\mathbf{b}_i = \frac{3}{4}\mathbf{n}_i$. Tetrahedral symmetry gives

$$\min_{\|\mathbf{a}\|_2=1} \sum_{i=1}^{4} |\mathbf{a}\cdot\mathbf{n}_i| = 2\sqrt{\frac{2}{3}}.$$

Thus its worst-case sensitivity is $\frac{3}{8}\,2\sqrt{2/3} = \sqrt{3/8}$, attaining the upper bound. This proves Eq. (20).

Theorem 7 (Finite reduction of the qutrit worst-case direction)**.** *Let $\mathbf{b}_1, \dots, \mathbf{b}_m \in \mathbb{R}^3$ span $\mathbb{R}^3$, and let*

$$h(\mathbf{a}) = \frac{1}{2}\sum_{k=1}^{m} |\mathbf{a}\cdot\mathbf{b}_k|, \qquad \|\mathbf{a}\|_2 = 1.$$

Then

$$\min_{\|\mathbf{a}\|_2=1} h(\mathbf{a}) = \min_{\substack{1\le i<j\le m \\ \mathbf{b}_i\times\mathbf{b}_j\neq 0}} h\!\left(\frac{\mathbf{b}_i\times\mathbf{b}_j}{\|\mathbf{b}_i\times\mathbf{b}_j\|_2}\right). \tag{21}$$

Repeated directions on the right-hand side may be removed. In particular, for a generic five-outcome rank-one qutrit POVM, the worst-case sensitivity is the minimum of at most ten explicitly computable facet values.

Proof. The function h is the support function of the centrally symmetric zonotope

$$Z = \sum_{k=1}^{m} [-\mathbf{b}_k/2, \mathbf{b}_k/2].$$

Since the generators span $\mathbb{R}^3$, the origin is an interior point of Z. The minimum of the support function over unit directions is therefore the radius of the largest Euclidean ball centered at the origin and contained in Z, equivalently the minimum distance from the origin to a facet of Z. Every facet of a three-dimensional zonotope is parallel to a two-dimensional span of generators; its normal is consequently parallel to $\mathbf{b}_i \times \mathbf{b}_j$ for some nonparallel pair. Evaluating the support function at these facet normals gives Eq. (21). If generators are in a degenerate configuration, the same statement follows by continuity, after omitting zero cross products and repeated normals.

Proposition 4 (An explicit five-outcome qutrit POVM)**.** *Let Q_0 be the 5×3 matrix*

$$Q_0 = \begin{pmatrix} \frac{1}{2\sqrt{2}} & -\frac{1+i\sqrt{3}}{4\sqrt{2}} & -\frac{1-i\sqrt{3}}{4\sqrt{2}} \\ \frac{1}{2\sqrt{2}} & -\frac{1+i\sqrt{3}}{4\sqrt{2}} & -\frac{1-i\sqrt{3}}{4\sqrt{2}} \\ \frac{1}{2} & \frac{1+i\sqrt{3}}{4} & \frac{1-i\sqrt{3}}{4} \\ \frac{1}{2} & \frac{1+i\sqrt{3}}{4} & \frac{-1+i\sqrt{3}}{4} \\ -\frac{1}{2} & \frac{1+i\sqrt{3}}{4} & \frac{-1+i\sqrt{3}}{4} \end{pmatrix}.$$

For an ordered pair of distinct indices $i, j \in \{1, \dots, 5\}$, let $G_{ij}(\theta, \phi)$ be the identity except for the block on the ordered coordinates (i, j), where it equals

$$\begin{pmatrix} \cos\theta & e^{i\phi}\sin\theta \\ -e^{-i\phi}\sin\theta & \cos\theta \end{pmatrix}.$$

Define

$$U = G_{34}\!\left(\frac{\pi}{41}, \frac{\pi}{2}\right) G_{41}\!\left(\frac{\pi}{66}, \pi\right) \times G_{23}\!\left(-\frac{\pi}{44}, 0\right) G_{25}\!\left(-\frac{\pi}{29}, \frac{3\pi}{2}\right), \qquad Q = UQ_0.$$

If q_k is the kth row of Q, then $E_k = q_k^\dagger q_k$ defines a rank-one five-outcome qutrit POVM. Its worst-case sensitivity is

$$\begin{aligned} 0.620509845202 &< \underline{\mathfrak{I}}_{\mathrm{meas}}(\mathsf{E}_5) \\ &< 0.620509845203. \end{aligned} \tag{22}$$

Thus five outcomes can strictly improve on the globally optimal four-outcome value.

Proof. Direct multiplication gives $Q_0^\dagger Q_0 = \mathbb{I}_3$, and every G_{ij} is unitary. Hence $Q^\dagger Q = \mathbb{I}_3$, so $\sum_k E_k = \mathbb{I}_3$ and each E_k is positive with rank one. In the Cartesian spin-1 representation, write

$$\mathbf{b}_k = \big(\mathrm{Tr}(E_k J_x), \mathrm{Tr}(E_k J_y), \mathrm{Tr}(E_k J_z)\big).$$

For a unit imaginary direction $\mathbf{a}$, the response is

$$R(\mathbf{a}) = \frac{1}{2}\sum_{k=1}^{5} |\mathbf{a}\cdot\mathbf{b}_k|.$$

This is the support function of the three-dimensional zonotope $\sum_k [-\mathbf{b}_k/2, \mathbf{b}_k/2]$. Its facets have normals parallel to $\mathbf{b}_i \times \mathbf{b}_j$. All ten cross products are nonzero. Their facet distances have outward interval enclosures listed in Supplemental Table S1, obtained from rational arithmetic, square roots, and trigonometric functions of rational multiples of π. The calculation uses `mpmath.iv` at 80 decimal digits, with outward-rounded endpoints checked again at the reported 12-decimal resolution. It separates the $(2, 3)$ facet from all the others and gives Eq. (22). In the same calculation,

$$\sqrt{3/8} < 0.612372435696,$$

so the improvement exceeds 0.0081374. Exact normalization follows analytically from $Q_0^\dagger Q_0 = \mathbb{I}$ and unitarity of the four factors; its interval enclosure is a separate consistency check. No global five-outcome optimality is claimed.

a. Computational verification. The supplied code separates ordinary 60-digit evaluation from outward interval certification. It also checks the Haar baseline and

compares the five-outcome facet formula with probabilities computed directly from density matrices. ChatGPT (OpenAI) assisted with restructuring, bilingual editing, and preparation of this verification code. The analytic arguments are stated explicitly; the calculations do not verify every cited source or every theorem. Additional rank-one and rank-two structural statements are supplied separately [22].

Theorem 8 (Exact sensitivity ranges of tetrahedral and icosahedral qutrit POVMs)**.** *Let $\mathbf{n}_j$ be the unit vertices of a regular tetrahedron and define $E_j^{\mathrm{tet}} = \frac{3}{4}P_{\mathbf{n}_j}$, where $P_{\mathbf{n}} = |\mathbf{n}\rangle\langle\mathbf{n}|$ is the spin-1 coherent projector. Then*

$$\underline{\mathfrak{I}}_{\mathrm{meas}}(\mathsf{E}_{\mathrm{tet}}) = \sqrt{\frac{3}{8}}, \qquad \mathfrak{I}_{\mathrm{meas}}(\mathsf{E}_{\mathrm{tet}}) = \frac{\sqrt{3}}{2}. \tag{23}$$

Let $\mathbf{n}_j$ be the 12 unit vertices of a regular icosahedron and define $E_j^{\mathrm{ico}} = \frac{1}{4}P_{\mathbf{n}_j}$. Then

$$\underline{\mathfrak{I}}_{\mathrm{meas}}(\mathsf{E}_{\mathrm{ico}}) = \sqrt{\frac{5+2\sqrt{5}}{20}}, \qquad \mathfrak{I}_{\mathrm{meas}}(\mathsf{E}_{\mathrm{ico}}) = \frac{1+\sqrt{5}}{4}. \tag{24}$$

Both are exact POVMs. Relative to the continuous qutrit optimum 3/4, their worst-case efficiencies are respectively $\sqrt{2/3}$ and $\sqrt{(20+8\sqrt{5})/45}$.

Proof. For a spin-1 coherent-state POVM with directions $\mathbf{n}_j$ and weights w_j, the ratio in a unit imaginary direction $\mathbf{a}$ is

$$R(\mathbf{a}) = \frac{3}{2}\sum_j w_j|\mathbf{a}\cdot\mathbf{n}_j|.$$

For the tetrahedron, $\sum_j \mathbf{n}_j = 0$ and $\frac{1}{4}\sum_j \mathbf{n}_j\mathbf{n}_j^T = \mathbb{I}_3/3$, so the four effects form a POVM. Its ratio is $R_{\mathrm{tet}}(\mathbf{a}) = \frac{3}{8}\sum_{j=1}^4 |\mathbf{a}\cdot\mathbf{n}_j|$, the support function of a four-generator zonotope. Its facet normals are perpendicular to pairs of tetrahedral vertices; tetrahedral symmetry makes all facet distances equal. Evaluating at $\mathbf{a} = (0,1,-1)/\sqrt{2}$ gives $\min R_{\mathrm{tet}} = \sqrt{3/8}$. For the maximum, use

$$\max_{\|\mathbf{a}\|=1}\sum_j |\mathbf{a}\cdot\mathbf{n}_j| = \max_{\mathbf{s}\in\{\pm1\}^4}\left\|\sum_j s_j\mathbf{n}_j\right\|.$$

The tetrahedral Gram matrix gives the maximal signed-sum norm $4/\sqrt{3}$, hence $\max R_{\mathrm{tet}} = \sqrt{3}/2$.

For the icosahedron, antipodal symmetry gives six unoriented axes $\mathbf{u}_1,\dots,\mathbf{u}_6$, and the ratio reduces to

$$R_{\mathrm{ico}}(\mathbf{a}) = \frac{1}{4}\sum_{j=1}^6 |\mathbf{a}\cdot\mathbf{u}_j|.$$

The 12 vertices satisfy $\sum_j \mathbf{n}_j = 0$ and $\frac{1}{12}\sum_j \mathbf{n}_j\mathbf{n}_j^T = \mathbb{I}_3/3$, so the effects again sum to $\mathbb{I}_3$. The zonotope generated by the six axes has facet normals parallel to $\mathbf{u}_i\times\mathbf{u}_j$. The 15 resulting unoriented normals form one orbit under the icosahedral symmetry group, so the support function has the same value at every facet normal. Taking $\mathbf{a} = \mathbf{e}_x$, with the standard vertex coordinates proportional to $(0,\pm1,\pm\varphi)$, $(\pm1,\pm\varphi,0)$, and $(\pm\varphi,0,\pm1)$, where $\varphi = (1+\sqrt{5})/2$, gives

$$\min R_{\mathrm{ico}} = \frac{\varphi^2}{2\sqrt{\varphi+2}} = \sqrt{\frac{5+2\sqrt{5}}{20}}.$$

Finally,

$$\max_{\|\mathbf{a}\|=1}\sum_{j=1}^6 |\mathbf{a}\cdot\mathbf{u}_j| = \max_{\mathbf{s}\in\{\pm1\}^6}\left\|\sum_j s_j\mathbf{u}_j\right\|.$$

Enumerating the 32 sign patterns modulo global sign, the possible squared norms are $6-2\sqrt{5}$, $6-6/\sqrt{5}$, $6+6/\sqrt{5}$, and $6+2\sqrt{5}$; the maximum is $6+2\sqrt{5} = 4\varphi^2$. Hence $\max R_{\mathrm{ico}} = \varphi/2 = (1+\sqrt{5})/4$. Dividing the two minimum values by 3/4 gives the stated relative efficiencies.

Theorem 9 (Dodecahedral 20-outcome qutrit POVM)**.** *Let $\mathbf{n}_j$, $j = 1,\dots,20$, be the unit vertices of a regular dodecahedron, and set $E_j^{\mathrm{dod}} = \frac{3}{20}P_{\mathbf{n}_j}$. Then $\mathsf{E}_{\mathrm{dod}}$ is a POVM and its exact worst-case sensitivity is*

$$\underline{\mathfrak{I}}_{\mathrm{meas}}(\mathsf{E}_{\mathrm{dod}}) = \frac{3(5+3\sqrt{5})}{20\sqrt{6}} \approx 0.716978. \tag{25}$$

Its worst-case efficiency relative to the continuous qutrit optimum 3/4 is

$$\frac{\underline{\mathfrak{I}}_{\mathrm{meas}}(\mathsf{E}_{\mathrm{dod}})}{3/4} = \frac{5+3\sqrt{5}}{5\sqrt{6}} \approx 95.60\%.$$

Proof. The 20 dodecahedral vertices have zero first moment and isotropic second moment, so $\sum_j E_j^{\mathrm{dod}} = \mathbb{I}_3$. Pair antipodal vertices to obtain ten unoriented axes $\mathbf{u}_1,\dots,\mathbf{u}_{10}$. The direction response is

$$R_{\mathrm{dod}}(\mathbf{a}) = \frac{3}{20}\sum_{j=1}^{10}|\mathbf{a}\cdot\mathbf{u}_j|.$$

The associated three-dimensional zonotope has a facet normal perpendicular to each pair of generating axes. The 45 pairwise cross products yield two symmetry classes: 30 normals have support value

$$\frac{3(5+3\sqrt{5})}{20\sqrt{6}},$$

and 15 normals have support value

$$\frac{3(2+\sqrt{5})}{10\sqrt{3}}.$$

These values follow by substituting the standard normalized dodecahedral coordinates $(\pm1,\pm1,\pm1)/\sqrt{3}$, $(0,\pm\varphi^{-1},\pm\varphi)/\sqrt{3}$, and their cyclic permutations, with $\varphi = (1+\sqrt{5})/2$. Since the minimum of a zonotope's support function on the unit sphere is its inradius, the smaller of the two facet distances gives Eq. (25).

VI. PEAK SENSITIVITY AND COHERENCE CONVERSION

A. Peak readout coefficient

A channel is called real here if it admits a Kraus representation with real matrix elements in the fixed reference basis. In a random mixture of POVMs, the classical label specifying the chosen POVM is retained as part of the outcome.

Theorem 10. *The measurement sensitivity satisfies* $0 \le \mathfrak{I}_{\mathrm{meas}}(\mathsf{E}) \le 1$. *It vanishes if and only if every event effect* $E(A)$ *is real in the reference basis (for a finite POVM, equivalently, every* E_k *is real). It is nonincreasing under classical postprocessing and under preprocessing by a real quantum channel, and it is convex under random mixing of POVMs.*

Proof. The upper bound follows from the data-processing inequality for trace distance:

$$\mathrm{TV}[p(\rho), p(\rho^*)] \le \frac{1}{2}\|\rho - \rho^*\|_1 = \mathcal{M}(\rho).$$

If every event effect $E(A)$ is real, then $\mathrm{Tr}[E(A)(\rho-\rho^*)] = 0$, because $\rho - \rho^*$ is purely imaginary and antisymmetric while $E(A)$ is real and symmetric. Hence $\mathfrak{I}_{\mathrm{meas}} = 0$. Conversely, if $\mathfrak{I}_{\mathrm{meas}} = 0$, then $\mathrm{Tr}[E(A)X] = 0$ for every measurable event A and every imaginary Hermitian direction X. Such directions span the full imaginary Hermitian sector, so the imaginary antisymmetric part of each event effect $E(A)$ must vanish. Classical postprocessing cannot increase total variation distance. For real preprocessing Λ, one has $\Lambda(\rho^*) = \Lambda(\rho)^*$ and $\mathcal{M}(\Lambda(\rho)) \le \mathcal{M}(\rho)$, which proves monotonicity. Convexity follows from convexity of total variation distance.

For a finite-outcome POVM, Eq. (5) can equivalently be written as

$$\mathfrak{I}_{\mathrm{meas}}(\mathsf{E}) = \sup_{\substack{X=X^\dagger,\ X^*=-X \\ X\neq 0}} \frac{\sum_k |\,\mathrm{Tr}(E_k X)|}{\|X\|_1}. \tag{26}$$

The admissible directions X are precisely the imaginary Hermitian directions; any such direction, after sufficiently small rescaling, can be realized as $\rho - \rho^*$ around the maximally mixed state.

Theorem 11 (Exact coarse-graining formula)**.** *Let* E *be a POVM on a measurable outcome space* (Ω, Σ)*, and let* $E(A)$ *be the effect associated with a measurable set* $A \in \Sigma$. *Then*

$$\mathfrak{I}_{\mathrm{meas}}(\mathsf{E}) = \sup_{A\in\Sigma} \|E(A) - E(A)^*\|_\infty. \tag{27}$$

For a finite-outcome POVM, this becomes

$$\mathfrak{I}_{\mathrm{meas}}(\mathsf{E}) = \max_{S\subseteq\{1,\dots,m\}} \|E_S - E_S^*\|_\infty, \qquad E_S = \sum_{k\in S} E_k.$$

In particular, for a binary POVM $\{E, \mathbb{I}-E\}$, $\mathfrak{I}_{\mathrm{meas}}(\mathsf{E}) = \|E - E^*\|_\infty$.

Proof. For probability measures, total variation distance is the supremum of the absolute probability difference over measurable events. Thus

$$\mathrm{TV}[p(\rho), p(\rho^*)] = \sup_{A\in\Sigma} |\,\mathrm{Tr}(E(A)X)|, \qquad X = \rho - \rho^*.$$

Since $\mathcal{M}(\rho) = \|X\|_1/2$, we may interchange the two suprema and optimize over imaginary Hermitian directions $X^* = -X$. For fixed A, only the imaginary part $B_A = (E(A) - E(A)^*)/2$ contributes to $\mathrm{Tr}(E(A)X)$. Trace/operator norm duality on this sector gives

$$\sup_{\substack{X=X^\dagger,\ X^*=-X \\ X\neq 0}} \frac{2|\,\mathrm{Tr}(E(A)X)|}{\|X\|_1} = 2\|B_A\|_\infty = \|E(A) - E(A)^*\|_\infty.$$

The upper bound follows from trace/operator norm duality. Equality is attained by a normalized difference of projectors onto conjugate eigenvectors associated with the eigenvalues $\pm\|B_A\|_\infty$. Taking the supremum over A proves Eq. (27). The finite-outcome formula follows by taking A to be each subset of outcomes.

B. Measurement-limited conversion

For this application only, fix an output basis and define the unnormalized trace-norm coherence

$$\mathcal{C}(\omega) = \min_{\delta\in\mathcal{I}} \|\omega - \delta\|_1, \tag{28}$$

where $\mathcal{I}$ is the set of diagonal output density operators. This convention is twice the normalized trace-distance-to-set quantity, as is Eq. (2).

Consider a measure-and-prepare protocol

$$\Lambda_{\mathsf{E}}(\rho) = \sum_k p_k(\rho)\tau_k, \tag{29}$$

where each τ_k is a fixed output state prepared after outcome k. For the resource-conversion interpretation, we take the prepared states to be real, so the apparatus does not inject additional imaginarity through state preparation. The inequality below in fact remains valid for arbitrary fixed output states. We quantify output coherence by Eq. (28) and define the imaginarity-attributable coherence change

$$\Delta\mathcal{C}_{\mathsf{E}}(\rho) = |\mathcal{C}[\Lambda_{\mathsf{E}}(\rho)] - \mathcal{C}[\Lambda_{\mathsf{E}}(\bar\rho)]|\,. \tag{30}$$

The baseline $\bar\rho$ removes the input's imaginary component while preserving its real part. This subtraction is essential: a real operation can generate real coherence even from a real input, so total output coherence alone is not a clean measure of conversion from imaginarity.

Theorem 12. *For every state* ρ*, POVM* E*, and measure-and-prepare protocol of the form* (29)*,*

$$\Delta\mathcal{C}_{\mathsf{E}}(\rho) \le \mathfrak{I}_{\mathrm{meas}}(\mathsf{E})\mathcal{M}(\rho). \tag{31}$$

Proof. Distance to a fixed set is 1-Lipschitz in the underlying norm, so

$$\Delta\mathcal{C}_{\mathsf{E}}(\rho) \le \|\Lambda_{\mathsf{E}}(\rho) - \Lambda_{\mathsf{E}}(\bar{\rho})\|_1.$$

Because $p_k(\bar{\rho}) = [p_k(\rho) + p_k(\rho^*)]/2$,

$$\begin{aligned}\|\Lambda_{\mathsf{E}}(\rho) - \Lambda_{\mathsf{E}}(\bar{\rho})\|_1 &= \left\| \frac{1}{2}\sum_k [p_k(\rho) - p_k(\rho^*)]\tau_k \right\|_1 \\ &\le \frac{1}{2}\sum_k |p_k(\rho) - p_k(\rho^*)| \\ &= \mathrm{TV}[p(\rho), p(\rho^*)] \\ &\le \mathfrak{I}_{\mathrm{meas}}(\mathsf{E})\mathcal{M}(\rho).\end{aligned}$$

Here $\|\tau_k\|_1 = 1$ for every output state. This proves Eq. (31).

A direct consequence is a no-go statement: if E is a real POVM, then $\mathfrak{I}_{\mathrm{meas}}(\mathsf{E}) = 0$ and $\Lambda_{\mathsf{E}}(\rho) = \Lambda_{\mathsf{E}}(\bar{\rho})$. Thus a real measurement followed only by outcome-dependent preparation cannot make the output depend on the imaginary component of the input. It may still produce coherence from the real component, but that coherence is present in the baseline as well.

C. State-adapted attainability

The upper bound is tight. Let $X = \rho - \rho^*$, and let P_+, P_-, and P_0 be the spectral projectors of X onto its positive, negative, and zero eigenspaces. Since $X^* = -X$, complex conjugation maps the positive and negative eigenspaces into one another, so $P_+^* = P_-$ and $P_0^* = P_0$. Define the two-outcome POVM

$$E_+ = P_+ + \frac{1}{2}P_0, \qquad E_- = P_- + \frac{1}{2}P_0. \tag{32}$$

It satisfies $E_+^* = E_-$ and is the Helstrom measurement for distinguishing ρ from ρ^*, with a symmetric split of the zero eigenspace.

Theorem 13. *For every finite-dimensional input ρ and every $s \in [0,1]$, there is a POVM $\mathsf{E}_\rho^{(s)}$ with $\mathfrak{I}_{\mathrm{meas}}(\mathsf{E}_\rho^{(s)}) = s$ and a real-output measure-and-prepare protocol for which*

$$\Delta\mathcal{C}_{\mathsf{E}^{(s)}}(\rho) = s\mathcal{M}(\rho). \tag{33}$$

Consequently, the maximum imaginarity-attributable trace-norm coherence at measurement sensitivity not exceeding s is exactly $s\mathcal{M}(\rho)$, when the measurement may be tailored to the known input state.

Proof. First suppose $\mathcal{M}(\rho) > 0$. The POVM in Eq. (32) obeys

$$p_+(\rho) - p_-(\rho) = \frac{1}{2}\|X\|_1 = \mathcal{M}(\rho),$$
$$p_+(\bar{\rho}) = p_-(\bar{\rho}) = \frac{1}{2}.$$

Prepare the real qubit states $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ for the $+$ and $-$ outcomes. The output is

$$\Lambda_{\mathsf{E}}(\rho) = \frac{1}{2}\left[\mathbb{I} + \mathcal{M}(\rho)\sigma_x\right], \qquad \Lambda_{\mathsf{E}}(\bar{\rho}) = \frac{\mathbb{I}}{2}.$$

Its trace-norm coherence is $\mathcal{M}(\rho)$, and the POVM has $\mathfrak{I}_{\mathrm{meas}}(\mathsf{E}) = 1$. Mixing it with the trivial unbiased measurement, $E_\pm^{(s)} = sE_\pm + (1-s)\mathbb{I}/2$, multiplies the sensitivity and output coherence by s, proving Eq. (33) in this case. If $\mathcal{M}(\rho) = 0$, then $\rho = \bar{\rho}$ and the left-hand side of Eq. (33) is zero. Choose instead the fixed POVM $E_\pm^{(s)} = [\mathbb{I} \pm sY_{01}]/2$, with $Y_{01} = -i|0\rangle\langle 1| + i|1\rangle\langle 0|$. It has sensitivity s and, for the real input, equal outcome probabilities; preparing $|+\rangle$ and $|-\rangle$ therefore yields zero output coherence. Thus Eq. (33) also holds when $\mathcal{M}(\rho) = 0$. The upper bound in Theorem 12 proves optimality.

The state-adapted construction above concerns exact coherence conversion for a known input; in general, the POVM $\mathsf{E}_\rho^{(s)}$ depends on ρ. It is distinct from the input-independent continuous POVM in Theorem 1, which optimizes the worst-case readout ratio but is not claimed to saturate the full coherence-conversion inequality for every state.

VII. QUBIT MEASUREMENTS: NOISE, MISALIGNMENT, AND SAMPLING

Here the number of shots refers to repeated uses of one fixed POVM. It is distinct from the number of outcomes optimized in Sec. IV.

Write a qubit state as

$$\rho = \frac{1}{2}(\mathbb{I} + r_x\sigma_x + r_y\sigma_y + r_z\sigma_z), \qquad r_x^2 + r_y^2 + r_z^2 \le 1. \tag{34}$$

Then $\mathcal{M}(\rho) = |r_y|$. A binary unsharp spin POVM is

$$E_\pm = \frac{1}{2}\left(\mathbb{I} \pm \eta\,\mathbf{n}\cdot\boldsymbol{\sigma}\right), \quad 0 \le \eta \le 1, \quad \|\mathbf{n}\| = 1. \tag{35}$$

Proposition 5. *For Eq. (35),*

$$\mathfrak{I}_{\mathrm{meas}}(\mathsf{E}) = \eta|n_y|. \tag{36}$$

If the outcomes prepare $|+\rangle$ and $|-\rangle$ in the output qubit, then

$$\Delta\mathcal{C}_{\mathsf{E}}(\rho) \le \eta|n_y|\,|r_y| = \mathfrak{I}_{\mathrm{meas}}(\mathsf{E})\mathcal{M}(\rho), \tag{37}$$

with equality for an aligned Y-axis measurement, $\mathbf{n} = \pm\hat{\mathbf{y}}$.

Proof. The outcome probabilities are $p_\pm = (1 \pm \eta\,\mathbf{n}\cdot\mathbf{r})/2$. Complex conjugation sends $r_y \mapsto -r_y$, giving $\mathrm{TV}[p(\rho), p(\rho^*)] = \eta|n_y r_y|$. Since $\mathcal{M}(\rho) = |r_y|$, Eq. (36) follows. The prepared output has the form $\frac{1}{2}[\mathbb{I} + \eta(\mathbf{n}\cdot\mathbf{r})\sigma_x]$. For $\bar{\rho}$, the corresponding Bloch coefficient is $\eta(n_x r_x + n_z r_z)$. The reverse triangle inequality gives Eq. (37).

Let δ be the angle between $\mathbf{n}$ and the Y axis, so $|n_y| = \cos\delta$ and $\sqrt{n_x^2 + n_z^2} = \sin\delta$ for $0 \le \delta \le \pi/2$. The sensitivity is $\eta\cos\delta$. Misalignment also creates a real-input baseline coherence bounded by

$$\mathcal{C}[\Lambda_{\mathsf{E}}(\bar{\rho})] \le \eta \sin\delta \sqrt{r_x^2 + r_z^2}. \tag{38}$$

Thus visibility loss reduces the attributable signal linearly, while axis misalignment both reduces sensitivity and permits false-positive coherence from a real input.

For illustration, the settings $(\eta, \delta) = (1, 15^\circ)$, $(0.8, 20^\circ)$, and $(0.5, 30^\circ)$ give sensitivity/leakage pairs $(0.966, 0.259)$, $(0.752, 0.274)$, and $(0.433, 0.250)$, respectively. Aligned settings have zero real-input leakage.

For an aligned unsharp Y measurement, each shot produces an outcome $Z_j \in \{+1, -1\}$ with mean $\mathbb{E}[Z_j] = \eta r_y$. The sample mean $\bar{Z}_N = N^{-1}\sum_j Z_j$ estimates the output coherence coefficient ηr_y. Hoeffding's inequality gives

$$\Pr\left(\left|\bar{Z}_N - \eta r_y\right| \ge \epsilon\right) \le 2\exp\left(-\frac{N\epsilon^2}{2}\right). \tag{39}$$

If η is calibrated and nonzero, $\widehat{\mathcal{M}} = |\bar{Z}_N|/\eta$ estimates $|r_y|$, with

$$\Pr\left(|\widehat{\mathcal{M}} - \mathcal{M}(\rho)| \ge \epsilon\right) \le 2\exp\left(-\frac{N\eta^2\epsilon^2}{2}\right).$$

The sample complexity therefore scales as η^{-2} for fixed confidence and accuracy. Misalignment requires an additional calibration or a real-input baseline measurement before interpreting the raw output coherence as imaginarity-derived.

VIII. DISCUSSION AND OUTLOOK

The central optimization is a lower measurement-norm constant on the imaginary Hermitian sector. Orthogonal two-frame covariance attains its exact dimension-dependent value. The complex Haar comparison makes the size of the improvement explicit: a factor $d/(d-1)$ at finite dimension, with the same leading asymptotic scaling. The four-outcome qutrit bound is global, while the five-outcome construction supplies a certified feasible improvement rather than a global optimizer.

Outcome complexity and copy complexity are separate. A positive worst-case coefficient requires at least $d(d-1)/2+1$ outcomes; the approximation construction matches that dimension exponent for fixed accuracy and confidence up to a logarithmic factor. The qutrit accuracy exponents remain nonmatching. None of these outcome statements is a finite-sample minimax estimator of arbitrary state imaginarity. Coherence conversion uses a peak coefficient and state-adapted attainability, with different quantifiers from the unknown-input guarantee.

The measurement-norm and zonoid framework is established [13, 14]; the exact restriction considered here and its attaining design should be assessed within that literature. Open problems include the unrestricted five-outcome optimum, matching qutrit accuracy exponents, statistical estimation with an experimentally specified outcome budget, and physical implementation costs. Those questions require further arguments beyond the coefficient computed here.

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DATA AVAILABILITY

The reproduction script and its numerical output for the finite-dimensional examples are included with the Supplemental Material [22]; no external datasets were used.

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[22] See Supplemental Material at [URL will be inserted by publisher] for the computational scripts, numerical data, and additional verification details.